

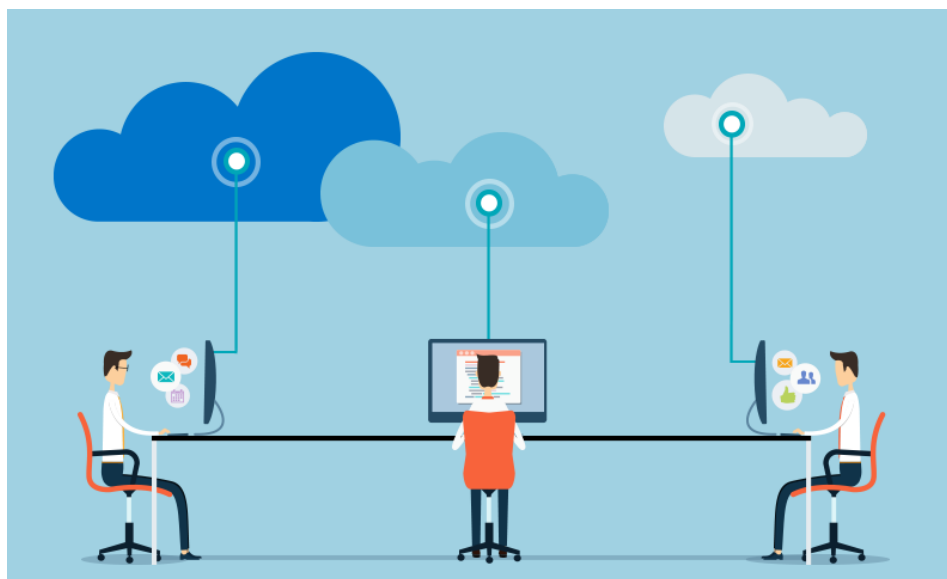


TECNOLÓGICO
NACIONAL DE MÉXICO



Tecnológico Nacional de México
Instituto Tecnológico de Saltillo

Departamento de Ingeniería en Sistemas Computacionales



Materia: Computo en la Nube

Profesor: Salazar Del Bosque Miguel

(7 – 8 hrs)

“Actividad 4.3 Seguridad”

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```

jesus@Debian3:~$ sudo apt install apache2 -y
apache2 is already the newest version (2.4.66-1~deb13u1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
jesus@Debian3:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-03-20 12:15:53 CST; 1h 1min ago
 Invocation: 4ea1c6a868204cb3b65ac3cc80a997d5
    Docs: https://httpd.apache.org/docs/2.4/
 Process: 844 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 948 (apache2)
   Tasks: 6 (limit: 3502)
  Memory: 21M (peak: 21.2M)
    CPU: 1.367s
   CGroup: /system.slice/apache2.service
           └─948 /usr/sbin/apache2 -k start
             └─952 /usr/sbin/apache2 -k start
               └─953 /usr/sbin/apache2 -k start
                 └─954 /usr/sbin/apache2 -k start
                   └─955 /usr/sbin/apache2 -k start
                     └─956 /usr/sbin/apache2 -k start

Mar 20 12:15:52 Debian3 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 20 12:15:53 Debian3 systemd[1]: Started apache2.service - The Apache HTTP Server.
jesus@Debian3:~$
jesus@Debian3:~$
jesus@Debian3:~$ ^C
jesus@Debian3:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
   qlen 1000
   link/ether 08:00:27:45:78:f6 brd ff:ff:ff:ff:ff:ff
   altname enx0800274578f6
   inet 192.168.56.120/24 brd 192.168.56.255 scope global enp0s8
       valid_lft forever preferred_lft forever
   inet6 fe80::a00:27ff:fe45:78f6/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether a6:60:cc:b1:20:bd brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
jesus@Debian3:~$

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jesus@Debian3:~$ cat /etc/apache2/conf-available/security.conf
# Changing the following options will not really affect the security of the
# server, but might make attacks slightly more difficult in some cases.

#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature Off

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On

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jesus@Debian3:~$
jesus@Debian3:~$ sudo a2enconf security
Conf security already enabled
jesus@Debian3:~$ sudo systemctl restart apache2
jesus@Debian3:~$ sudo a2dismod autoindex status proxy proxy_http
WARNING: The following essential module will be disabled.
This might result in unexpected behavior and should NOT be done
unless you know exactly what you are doing!
autoindex

To continue type in the phrase 'Yes, do as I say!' or retry by passing '-f': Yes, do as I say!
Module autoindex disabled.
Module status disabled.
Module proxy already disabled
Module proxy_http already disabled
To activate the new configuration, you need to run:
systemctl restart apache2
jesus@Debian3:~$ sudo systemctl restart apache2
jesus@Debian3:~$

```




debian

Apache2 Debian Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining

```

jesus@Debian3:~$ sudo apache2ctl configtest
Syntax OK
jesus@Debian3:~$ sudo systemctl restart apache2
jesus@Debian3:~$
jesus@Debian3:~$ sudo ss -tlnl
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port
udp        UNCONN     0           0            0.0.0.0:5353             0.0.0.0:*
udp        UNCONN     0           0            0.0.0.0:48421            0.0.0.0:*
udp        UNCONN     0           0            [::]:44259              [::]:*
udp        UNCONN     0           0            [::]:5353               [::]:*
tcp        LISTEN     0           80          127.0.0.1:3306           0.0.0.0:*
tcp        LISTEN     0           128         0.0.0.0:22              0.0.0.0:*
tcp        LISTEN     0          4096        127.0.0.1:631           0.0.0.0:*
tcp        LISTEN     0           511         *:443                   *:443
tcp        LISTEN     0           128         [::]:22                 [::]:*
tcp        LISTEN     0          4096        [::1]:631               [::]:*
tcp        LISTEN     0           511         *:80                    *:80
jesus@Debian3:~$ sudo ufw allow 80
Rules updated
Rules updated (v6)
jesus@Debian3:~$ sudo ufw allow 443
Rules updated
Rules updated (v6)
jesus@Debian3:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y/n)? y
Firewall is active and enabled on system startup
jesus@Debian3:~$ sudo tail -f /var/log/apache2/access.log
192.168.56.1 - - [20/Mar/2026:13:38:05 -0600] "GET / HTTP/1.1" 200 3367 "-" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:13:38:07 -0600] "GET / HTTP/1.1" 200 3366 "-" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:13:38:07 -0600] "GET / HTTP/1.1" 200 3366 "-" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
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192.168.56.1 - - [20/Mar/2026:13:38:07 -0600] "GET / HTTP/1.1" 200 3366 "-" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:13:38:25 -0600] "GET / HTTP/1.1" 200 6243 "-" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:13:38:25 -0600] "GET /icons/openlogo-75.png HTTP/1.1" 200 6046 "https://192.168.56.120/" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:13:38:25 -0600] "GET /favicon.ico HTTP/1.1" 404 458 "https://192.168.56.120/" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:13:38:55 -0600] "GET / HTTP/1.1" 200 4796 "-" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:13:38:55 -0600] "GET /icons/openlogo-75.png HTTP/1.1" 200 6046 "https://192.168.56.120/" Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"

```

```

jesus@Debian3:~$ sudo tail -f /var/log/apache2/error.log
[Fri Mar 20 13:24:41.913337 2026] [mpm_prefork:notice] [pid 1820:tid 1820] AH00170: caught SIGWINCH, s
hutting down gracefully
[Fri Mar 20 13:24:42.130162 2026] [mpm_prefork:notice] [pid 1857:tid 1857] AH00163: Apache/2.4.66 (Deb
ian) configured -- resuming normal operations
[Fri Mar 20 13:24:42.131794 2026] [core:notice] [pid 1857:tid 1857] AH00094: Command line: '/usr/sbin/
apache2'
[Fri Mar 20 13:37:05.209113 2026] [mpm_prefork:notice] [pid 1857:tid 1857] AH00170: caught SIGWINCH, s
hutting down gracefully
[Fri Mar 20 13:37:05.370431 2026] [ssl:warn] [pid 1996:tid 1996] AH01906: Debian3.0.myquest.virtualbox
.org:443:0 server certificate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Mar 20 13:37:05.370525 2026] [ssl:warn] [pid 1996:tid 1996] AH01909: Debian3.0.myquest.virtualbox
.org:443:0 server certificate does NOT include an ID which matches the server name
[Fri Mar 20 13:37:05.427020 2026] [ssl:warn] [pid 1997:tid 1997] AH01906: Debian3.0.myquest.virtualbox
.org:443:0 server certificate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Mar 20 13:37:05.427067 2026] [ssl:warn] [pid 1997:tid 1997] AH01909: Debian3.0.myquest.virtualbox
.org:443:0 server certificate does NOT include an ID which matches the server name
[Fri Mar 20 13:37:05.441026 2026] [mpm_prefork:notice] [pid 1997:tid 1997] AH00163: Apache/2.4.66 (Deb
ian) OpenSSL/3.5.4 configured -- resuming normal operations
[Fri Mar 20 13:37:05.441087 2026] [core:notice] [pid 1997:tid 1997] AH00094: Command line: '/usr/sbin/
apache2'

^C
jesus@Debian3:~$
jesus@Debian3:~$

```